

Ethan Buckley

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Education

University College London, MSci Natural Sciences

Sep 2024 to Jun 2028

- Physics major, Physical Chemistry minor; predicted First-Class Honours
- Relevant modules: Mathematical Methods III (PDEs and Fourier analysis), Quantum Physics, Thermal and Statistical Physics
- Extracurriculars: AI Society, Natural Sciences Society

Experience

Data Science and Research Intern, Down Syndrome Education International

Jun 2026 to present

- Build hierarchical Bayesian models (PyMC, NUTS) of vocabulary growth in children with Down syndrome: Beta-Binomial likelihoods, Gaussian-process age trajectories and study- and child-level random effects, fitted to 1,424 assessments pooled from 14 datasets in six countries.
- Wrote the intention-to-treat, difference-in-differences and mediation models for a Bayesian re-analysis of a randomised reading trial (54 children, four waves), inside a causal DAG with LightGBM and SHAP predictor ranking.
- Built the pipeline that generates the charity's teaching images, audio and video from a written specification, and the audit, repair and upload route that files the results as draft assets in its catalogue. Its main rule is figure-ground contrast, for the reduced contrast sensitivity common in Down syndrome.
- Chose text-to-speech voices with a blind A/B listening arena, Bradley-Terry ranking and bootstrap intervals over four commercial providers; found and fixed a scoring defect that awarded first place by sort order on tied comparisons.
- Prior and posterior predictive checks, convergence gates (R -hat at most 1.01, effective sample size at least 400) and PSIS-LOO and leave-one-study-out model comparison, all run in CI.

Literature review, UCL Centre for Computational Science

From Oct 2026

- Literature review on quantum-GPU computing and variational quantum algorithms for biomolecular simulation, supervised by Professor Peter Coveney FRS. Begins October 2026.

Facilities Team Member, Roadchef, Killington Lake

Jul 2025 to Apr 2026

- Facilities team for a motorway services: site cleaning and maintenance, taking in and distributing deliveries for every outlet, and regular cover at Costa, WHSmith and Days Inn housekeeping.

Placements, BAE Systems (Barrow) and EDF Energy (Heysham 1)

Jul to Aug 2023

- Placements across engineering, chemistry and maintenance in safety-critical nuclear sites, including reactor-chemistry monitoring in the Heysham laboratory.

Projects

- **S&P 500 stock screener (independent):** XGBoost classifier on about 1.3 million daily observations plus FinBERT news sentiment; 22 technical indicators; triple-barrier labelling; walk-forward validation over 1,630 out-of-sample days (2020 to 2026), top-15 precision about 48% against a 29% base rate. Survivorship bias and transaction costs reported as limitations.
- **SEIQR epidemic cellular automaton (UCL group project, individually extended):** Vectorised simulation (about 40-fold faster), validated against the mean-field ODE limit, bit-identical regression tests in CI on Python 3.10 to 3.12, Streamlit dashboard.
- **The Computable World (independent):** A referenced investment thesis on AI, virtual twins, robotics and the industrial stack behind them, published at ethanbuckley.github.io/computable-world.

Relevant Skills

- **Programming:** Python (NumPy, pandas, PyMC, XGBoost, LightGBM, SHAP, scikit-learn, pytest), SQL and DuckDB, Git and GitHub Actions, Quarto, LaTeX, generative-model APIs with batch inference.
- **Statistics and machine learning:** Bayesian inference and MCMC, hierarchical models, Gaussian processes, gradient boosting, causal inference (DAGs, ITT, difference-in-differences, mediation), blind A/B evaluation with Bradley-Terry ranking, NLP with FinBERT.

Extracurricular and Interests

- UCL Men's Hockey Club: play for the club.
- Volunteer junior coach, Kirkby Lonsdale Hockey Club.
- Interests: mountaineering, skiing, running.